COMPLETING A BLOCK QUADRATIC FORM

The question is whether a coupled quadratic form is strictly positive. We eliminate its mixed term, read off the residual form, and then use the same calculation to minimize over one block of variables.

1. A LOCAL POSITIVITY PROBLEM

The reader needs only matrix multiplication, quadratic-form positivity and an invertible change of variables. The Schur complement will emerge from the calculation rather than serve as an assumed rule.

Theorem 1.1. *Let $p, q \geq 1$, let $A \in \mathbb{R}^{p\times p}$ and $C \in \mathbb{R}^{q\times q}$ be symmetric, and let $B \in \mathbb{R}^{p\times q}$. Assume $A > 0$, where $H > 0$ means $v^T H v > 0$ for every nonzero real vector v of the appropriate size. Set*

$$M = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix}, \qquad S = C - B^T A^{-1} B.$$

Then $M > 0$ if and only if $S > 0$. Moreover, for every fixed $y \in \mathbb{R}^q$,

$$\min_{x\in\mathbb{R}^p} (x, y)^T M (x, y) = y^T S y, \qquad x_{\min} = -A^{-1} B y.$$

The minimizer is unique. This is a classical quadratic-form identity and criterion.

The point: cancel the mixed term. For fixed blocks in the theorem, put

$$Q(x, y) = x^T A x + 2x^T B y + y^T C y.$$

The mixed term will occur in a square if the new variable z satisfies $Az = Ax + By$. Since $A > 0$ implies that A is invertible, choose

$$z = x + A^{-1} B y, \qquad x = z - A^{-1} B y. \tag{1.1}$$

Symmetry gives $(A^{-1})^T = A^{-1}$. Expanding without discarding any term,

$$z^T A z = x^T A x + 2x^T B y + y^T B^T A^{-1} B y.$$

Subtracting the last term from the y-part produces

$$Q(x, y) = z^T A z + y^T \underbrace{(C - B^T A^{-1} B)}_{S} y. \tag{1.2}$$

The matrix S is called the Schur complement of A. Its role is now visible: it is exactly the residual quadratic form after cancellation of the coupling; see [1, Appendix A.5.5].

The connection: both implications. The two formulas in (1.1) show that $(x, y) \mapsto (z, y)$ is a bijection preserving the nonzero-pair condition. If $S > 0$, each nonzero pair has a nonzero component, so (1.2) is strictly positive.

Conversely, suppose $M > 0$. For each $y \neq 0$, take $x = -A^{-1}By$. Then $(x, y) \neq 0$ while $z = 0$, and

$$0 < Q(-A^{-1}By, y) = y^T Sy.$$

This proves $S > 0$ because the test applies to every nonzero y. The argument uses positivity of a quadratic form, without assuming that the entire block matrix is invertible in advance.

2. The minimum and the limit of the argument

For a fixed y, the first term of (1.2) is nonnegative and vanishes only when $z = 0$. The inverse change therefore gives the unique minimum in Theorem 1.1.

For the scalar blocks $A = 2, B = 1, C = 1$, the calculation is concrete:

$$Q(x, y) = 2x^2 + 2xy + y^2 = 2\left(x + \frac{y}{2}\right)^2 + \frac{y^2}{2}.$$

Thus $S = 1/2 > 0$, and the minimizing choice is $x = -y/2$. Both positivity and the minimum value can be checked in the same displayed expression.

If A is symmetric and merely invertible, the square identity still holds. The inequality needed to minimize need not hold. For example,

$$A = -1, \quad B = 0, \quad C = 1 \quad \Longrightarrow \quad S = 1, \quad Q(x, y) = -x^2 + y^2.$$

Here S is positive but M is indefinite, and $Q(\cdot, y)$ is unbounded below. If A is singular, A^{-1} is unavailable; generalized criteria require additional work and are not asserted here.

References

[1] Stephen Boyd and Lieven Vandenberghe. Convex optimization. Online source, 2004. published; controlling version: 2004; author-hosted PDF accessed 2026-10-12.